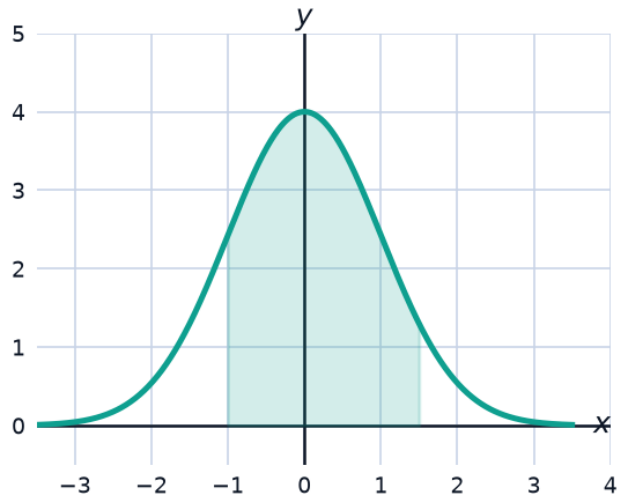
Neural Layer Matrices and Activations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

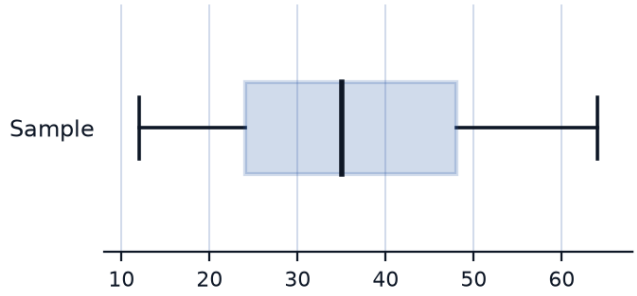
1 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Consider a dataset with n data points $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, where each $\mathbf{x}_i \in \mathbb{R}^p$. The sample covariance matrix is given by $\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$, where $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i$ is the sample mean. Suppose PCA is performed on this dataset, resulting in principal components defined by the eigenvectors of $\mathbf{S}$ and corresponding variances given by the eigenvalues. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ be the orthonormal eigenvectors of $\mathbf{S}$ with associated eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$. The first principal component captures the direction of maximum variance, λ_1, and the second principal component captures the next largest variance, λ_2, orthogonal to the first. If we decide to reduce the dimensionality from p to k by keeping only the first k principal components, the proportion of the total variance retained is given by $\frac{\sum_{i=1}^{k}\lambda_i}{\sum_{i=1}^{p}\lambda_i}$. Suppose for a particular dataset, the eigenvalues of the covariance matrix are $\{10.5, 5.2, 2.1, 0.8, 0.4\}$. If we aim to retain at least 90% of the total variance, what is the minimum number of principal components required?



2 [Foundational] • Target: Affine Transformations and Matrix Dimensions

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 62$, $\lambda_2 = 24$, and $\lambda_3 = 14$. What percentage of total variance is explained by the first principal component PC_1?



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3 [Foundational] • Target: Neural Parameter Counting

A dataset consists of three independent observations X_1, X_2, X_3 drawn from a Poisson distribution with parameter λ. The observed values are $x_1 = 3, x_2 = 5, x_3 = 4$. We want to find the Maximum Likelihood Estimate (MLE) for λ. The log-likelihood function for a Poisson distribution is given by $\ell(\lambda) = \sum_{i=1}^{n} (x_i \log(\lambda) - \log(x_i!) - \lambda)$. Part 1: Derive the expression for the derivative of the log-likelihood function with respect to λ. Part 2: Set the derivative to zero to find the MLE of λ.

4 [Foundational] • Target: Rectified Activations and Dying Neurons

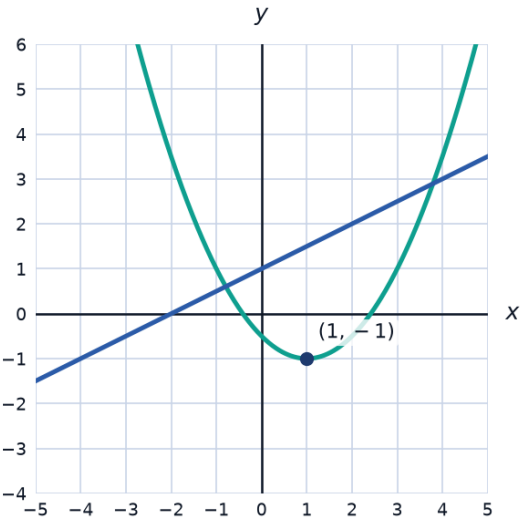
A dataset consists of $n = 100$ independent and identically distributed (i.i.d.) observations $x_1, x_2, \ldots, x_{100}$. These observations are drawn from a Bernoulli distribution with parameter p. The probability mass function (PMF) of a Bernoulli distribution is given by $P(X = x; p) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$. The likelihood function $L(p)$ for this dataset is the product of the PMFs of the individual observations. If we observe $k = 60$ successes (i.e., $x_i = 1$) and $n - k = 40$ failures (i.e., $x_i = 0$), what is the likelihood function $L(p)$ for p?

5 [Foundational] • Target: Saturating Activations and Softmax Probabilities

Consider a dataset with two features, X_1 and X_2. The covariance matrix for this dataset is given by: $\Sigma = (10, 5; 5, 20)$ What is the variance of the first feature, X_1?

6 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 70$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.



7 [Foundational] • Target: Affine Transformations and Matrix Dimensions

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{50}$ to $\mathbf{y} \in \mathbb{R}^{25}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Neural Parameter Counting**

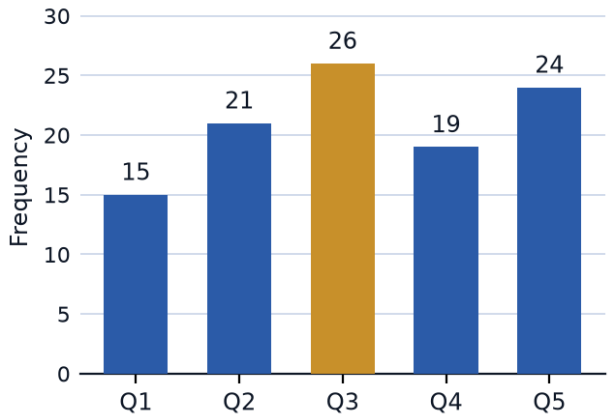
Consider a dataset where the number of defective items produced by a machine per hour, Y, follows a Poisson distribution with a rate parameter $\lambda = 3$. The probability mass function (PMF) for a Poisson distribution is given by $P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \ldots$. What is the value of λ in this scenario?

9 **[Intermediate]** • **Target: Rectified Activations and Dying Neurons**

Consider two events, A and B, with the following probabilities: $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. What is the conditional probability of A given B, denoted as $P(A|B)$?

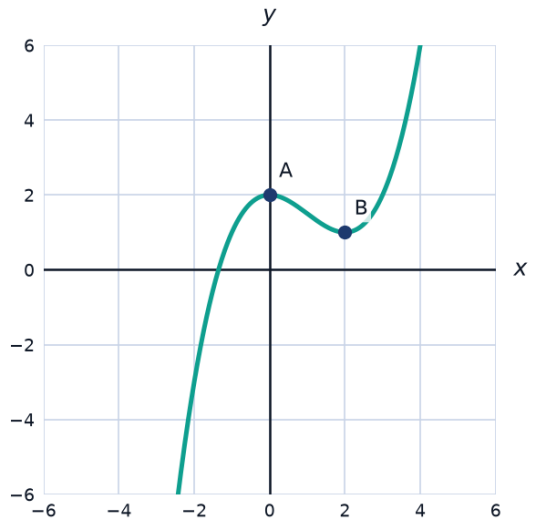
10 **[Intermediate]** • **Target: Saturating Activations and Softmax Probabilities**

A centered feature matrix $\mathbf{X}$ has shape $N \times 22$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?



11 **[Intermediate]** • **Target: Neural Layer Matrices & Activations Practice Exercises**

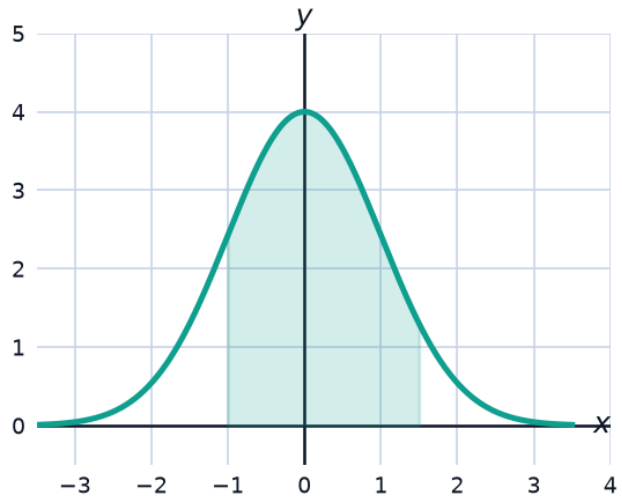
Consider the scalar loss function $L(w) = 6w^2 - 8w + 5$. Find the analytical gradient $ablaL(w)$ evaluated at $w = 10$.



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12 [Intermediate] • Target: Affine Transformations and Matrix Dimensions

Given $n = 12$ independent observations from a normal distribution with sum $\sum_{i=1}^{12} x_i = 36$, what is the Maximum Likelihood Estimate (MLE) $\hat{\mu}_{\text{MLE}}$?



13 [Intermediate] • Target: Neural Parameter Counting

A medical diagnostic test for a rare disease is designed to be highly sensitive and specific. The disease affects 1 in every 1000 individuals in the population. The test has a sensitivity of 99% (meaning it correctly identifies 99% of individuals who have the disease) and a specificity of 95% (meaning it correctly identifies 95% of individuals who do not have the disease). If a randomly selected individual from the population tests positive, what is the probability that they actually have the disease?

14 [Intermediate] • Target: Rectified Activations and Dying Neurons

Consider a dataset of n data points, each residing in $\mathbb{R}^D$. We are performing Principal Component Analysis (PCA) to reduce the dimensionality. The first three principal components capture 70% of the total variance, the first six capture 85%, and the first ten capture 92%. You are tasked with selecting a subset of principal components for a downstream machine learning model. If the model requires that at least 90% of the original variance be retained, and you want to use the minimum possible number of principal components, what is the dimensionality of the transformed data?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

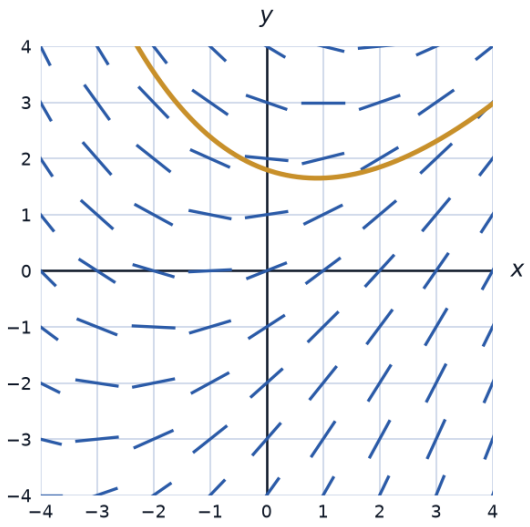
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15 [Challenging] • Target: Saturating Activations and Softmax Probabilities • Cross-Lesson: 3.01 Gradient Descent and Optimization

A medical test for a rare disease has a 99\% accuracy rate in detecting the disease when it is present (True Positive Rate) and a 95\% accuracy rate in identifying individuals without the disease (True Negative Rate). If the prevalence of the disease in the population is 0.5\%, what is the probability that a person who tests positive actually has the disease?

16 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider a binary classification problem where the decision boundary is defined by the logistic function $p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$, which represents the probability of a data point $\mathbf{x} = (x_1, x_2)$ belonging to class 1. The estimated model parameters are $\beta_0 = -3$, $\beta_1 = 1$, and $\beta_2 = -2$. We are interested in the geometric shape of the classification boundary. The classification boundary occurs where $p(x) = 0.5$. Find the equation of this boundary in the x_1x_2-plane.



17 [Challenging] • Target: Affine Transformations and Matrix Dimensions • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Consider the quadratic loss function $L(w) = w^2 - 4w + 3$. Find the value of w where the function reaches its minimum.

18 [Challenging] • Target: Neural Parameter Counting • Cross-Lesson: 6.01 Linear Regression and OLS

Consider the following quadratic loss function: $L(w) = 3w^2 - 12w + 10$. What is the value of w that minimizes this loss function?

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19 [Challenging] • Target: Rectified Activations and Dying Neurons • Cross-Lesson: 7.01 Logistic Regression and Classification

Consider a simple quadratic loss function $L(w) = (w^2 - 4w + 5)$. We want to find the minimum of this loss function using gradient descent. The gradient of the loss function with respect to w is given by $\frac{dL}{dw} = 2w - 4$. If we start at an initial weight $w_0 = 5$ and use a learning rate $\alpha = 0.5$, what will be the value of the weight w_1 after the first update step, i.e., $w_1 = w_0 - \alpha \frac{dL}{dw}(w_0)$?

20 [Challenging] • Target: Saturating Activations and Softmax Probabilities • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -4$ and $z_2 = 6$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input \(\mathbf{x} \in \mathbb{R}^{10} \) to \(\mathbf{y} \in \mathbb{R}^{5} \) using \(\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b} \). How many total learnable parameters are in this layer?

a. \(55 \)

b. \(50 \)

c. \(15 \)

d. \(65 \)

22 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input \(\mathbf{x} \in \mathbb{R}^{20} \) to \(\mathbf{y} \in \mathbb{R}^{10} \) using \(\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b} \). How many total learnable parameters are in this layer?

a. \(30 \)

b. \(200 \)

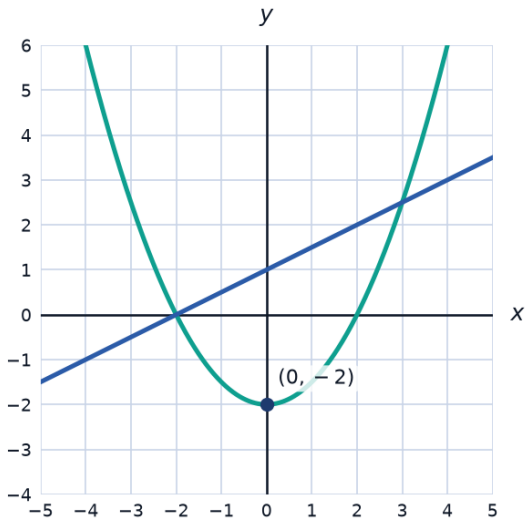
c. \(210 \)

d. \(230 \)

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23 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{30}$ to $\mathbf{y} \in \mathbb{R}^{15}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?



a. 45

b. 450

c. 465

d. 495

24 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{40}$ to $\mathbf{y} \in \mathbb{R}^{20}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

a. 60

b. 800

c. 820

d. 860

25 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{50}$ to $\mathbf{y} \in \mathbb{R}^{25}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

a. 1275

b. 1250

c. 75

d. 1325

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26 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

A fully-connected dense layer maps an input $\mathbf{x} \in \mathbb{R}^{60}$ to $\mathbf{y} \in \mathbb{R}^{30}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

a. 1830

b. 1800

c. 90

d. 1890

27 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -2$ and $z_2 = 3$.

a. $f(z_1) = 2, \; f(z_2) = 3$

b. $f(z_1) = -2, \; f(z_2) = 3$

c. $f(z_1) = 0, \; f(z_2) = 0$

d. $f(z_1) = 0, \; f(z_2) = 3$

28 [Foundational] • Target: Neural Layer Matrices & Activations Practice Exercises

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -4$ and $z_2 = 6$.

a. $f(z_1) = 4, \; f(z_2) = 6$

b. $f(z_1) = -4, \; f(z_2) = 6$

c. $f(z_1) = 0, \; f(z_2) = 0$

d. $f(z_1) = 0, \; f(z_2) = 6$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Affine Transformations and Matrix Dimensions

A fully-connected neural layer maps input $\mathbf{x} \in \mathbf{R}^{128}$ to output $\mathbf{y} \in \mathbf{R}^{64}$ via $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. What is the shape of the weight matrix $\mathbf{W}$?

a. 64×128

b. 128×64

c. 128×128

d. 64×64

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30 [Intermediate] • Target: Affine Transformations and Matrix Dimensions

For a mini-batch of $B = 32$ samples with input dimension $D = 128$, what is the shape of the activation batch $\mathbf{Y} = \mathbf{X}\mathbf{W}^T$ when $\mathbf{W}$ is 64×128?

a. 32×64

b. 64×32

c. 32×128

d. 128×64

31 [Intermediate] • Target: Affine Transformations and Matrix Dimensions

If a two-layer linear network computes $\mathbf{y} = \mathbf{W}_2(\mathbf{W}_1\mathbf{x})$ without non-linear activations, why is its expressive capacity limited to a single linear transformation?

a. The product of two matrices $\mathbf{W}_3 = \mathbf{W}_2\mathbf{W}_1$ is itself a single matrix

b. The gradient of two matrices is always undefined

c. Matrix multiplication is non-associative

d. The bias vector becomes zero

32 [Intermediate] • Target: Neural Parameter Counting

Calculate the total number of trainable parameters (weights + biases) in a dense layer with 256 inputs and 128 outputs.

a. 32928

b. 32768

c. 33024

d. 32896

33 [Intermediate] • Target: Neural Parameter Counting

A two-layer perceptron has architecture $10 \to 20 \to 1$. How many total parameters are in the network (including all biases)?

a. 241

b. 220

c. 200

d. 231

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34 **[Intermediate]** • **Target: Neural Parameter Counting**

A convolutional layer has 32 filters of size 3×3 across 3 input channels with a bias per filter. How many learnable parameters does it contain?

a. 896

b. 864

c. 288

d. 928

35 **[Intermediate]** • **Target: Rectified Activations and Dying Neurons**

Evaluate the standard ReLU activation function $\mathrm{ReLU}(z) = \max(0, z)$ and its derivative for $z = -3.5$.

a. $\mathrm{ReLU}(-3.5) = 0, \quad \mathrm{ReLU}'(-3.5) = 0$

b. $\mathrm{ReLU}(-3.5) = -3.5, \quad \mathrm{ReLU}'(-3.5) = 1$

c. $\mathrm{ReLU}(-3.5) = 0, \quad \mathrm{ReLU}'(-3.5) = 1$

d. $\mathrm{ReLU}(-3.5) = 3.5, \quad \mathrm{ReLU}'(-3.5) = -1$

36 **[Intermediate]** • **Target: Rectified Activations and Dying Neurons**

How does the Leaky ReLU activation $f(z) = \max(0.01z, z)$ mitigate the 'dying ReLU' problem during backpropagation?

a. It provides a non-zero gradient (0.01) for negative inputs, preventing weights from freezing

b. It normalizes inputs to have zero mean and unit variance

c. It bounds outputs between -1 and 1

d. It eliminates the need for bias vectors

37 **[Intermediate]** • **Target: Rectified Activations and Dying Neurons**

Find the output of a Leaky ReLU activation with coefficient $\alpha = 0.1$ when input is $z = -8$.

a. -0.8

b. 0

c. -8.0

d. 0.8

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Saturating Activations and Softmax Probabilities • Cross-Lesson: 6.01 Linear Regression and OLS

The standard logistic sigmoid is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.8$, compute the derivative $\sigma'(z_0)$.

a. 0.16

b. 0.80

c. 0.64

d. 0.20

39 [Challenging] • Target: Saturating Activations and Softmax Probabilities • Cross-Lesson: 7.01 Logistic Regression and Classification

Why does the Hyperbolic Tangent activation $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$ generally train faster than standard sigmoid in hidden layers?

a. It is zero-centered with outputs in $(-1, 1)$, preventing systematic gradient bias

b. Its derivative is identically 1 everywhere

c. It does not require exponent calculation

d. It eliminates all negative activations

40 [Challenging] • Target: Saturating Activations and Softmax Probabilities • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Apply the Softmax function $\text{softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum e^{z_j}}$ to the logit vector $\mathbf{z} = [0, \ln 2, \ln 5]^T$. Find the probability of class 3.

a. $\frac{5}{8} = 0.625$

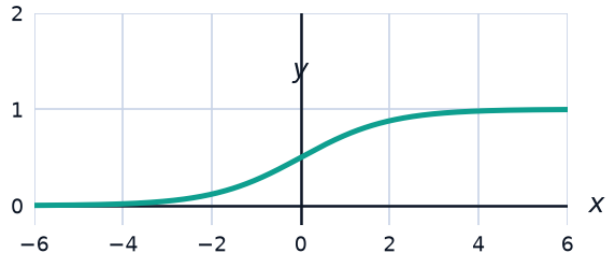
b. $\frac{1}{8} = 0.125$

c. $\frac{2}{8} = 0.250$

d. $\frac{5}{7} \approx 0.714$

41 [Challenging] • Target: Neural Layer Matrices & Activations Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.20$, what is the value of the derivative $\sigma'(z_0)$?



a. 0.1600

b. 0.4000

c. 0.8000

d. 0.0400

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42 [Challenging] • Target: Affine Transformations and Matrix Dimensions • Cross-Lesson: 3.01 Gradient Descent and Optimization

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 1.20$, what is the value of the derivative $\sigma'(z_0)$?

a. 2.4000

b. -0.2400

c. -0.2000

d. 1.4400

43 [Challenging] • Target: Neural Parameter Counting • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.40$, what is the value of the derivative $\sigma'(z_0)$?

a. 0.2400

b. 0.8000

c. 0.6000

d. 0.1600

44 [Challenging] • Target: Rectified Activations and Dying Neurons • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.60$, what is the value of the derivative $\sigma'(z_0)$?

a. 0.2400

b. 1.2000

c. 0.4000

d. 0.3600

45 [Challenging] • Target: Saturating Activations and Softmax Probabilities • Cross-Lesson: 6.01 Linear Regression and OLS

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.80$, what is the value of the derivative $\sigma'(z_0)$?

a. 0.1600

b. 1.6000

c. 0.2000

d. 0.6400

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